

Project title: FB15k-237 Link Prediction: Baseline Training and Preliminary Experiments **Group:** Thalia Delhaise, Lenny Briclet, André Fonseca, Rafael Gufler

Overall assessment

The current status is promising. The group has a well-defined task, a standard benchmark dataset, a valid evaluation protocol, two implemented baselines, preliminary results, and training curves. The project is already technically de-risked. The main remaining challenge is to turn the planned analysis into a clear scientific contribution, rather than only reporting that RotatE outperforms TransE.

What is already working

- The link prediction task is precisely defined: predict missing heads or tails in FB15k-237 triples.
- The dataset is standard, already loaded, and uses the official train/validation/test split.
- The evaluation protocol is appropriate: filtered MRR and Hits@k.
- Two real baselines are implemented: TransE and RotatE.
- Preliminary results are reported and interpreted.
- Training curves are included and show that both models were still improving.
- The next-step plan is concrete: hard negative mining, slice metrics, and error analysis.

Main issues to address

- The central claim about relation-structural patterns still needs operational definitions.
- The comparison is not yet fully conclusive because the models were trained for only 15 epochs.
- “Matched hyperparameters” should be described more carefully, since TransE and RotatE may use different default losses or model-specific settings in PyKEEN.
- The hard-negative mining contribution should be kept small and testable.
- Slice evaluation must avoid leakage: relation degrees and structural categories should be computed from the training graph only.
- Error analysis is planned but not yet shown.

Detailed comments

1. Problem and evaluation

The problem is very clear. The report states the task, dataset, graph formalism, and ranking protocol well. Using FB15k-237 is appropriate because it is a

standard benchmark for multi-relational link prediction.

The evaluation metrics are also appropriate. Filtered MRR and Hits@1/3/10 are the right choices for this task. For Milestone 2, the report should explicitly state whether metrics are macro-averaged over head/tail prediction or computed over all queries jointly. Since you already observe a strong head-tail asymmetry, this distinction matters.

2. Data pipeline

The data pipeline is simple but solid. Loading FB15k-237 through PyKEEN is acceptable, provided that the exact dataset version, library version, random seed, and generated artifacts are documented. The report already mentions cached loading, entity/relation ID mapping, triple count checks, and saved graph statistics.

For the next version, include a small table with the final dataset statistics: number of entities, relations, train/validation/test triples, average degree, and relation-frequency range. This will support the later slice analysis.

3. Evaluation validity

The filtered evaluation protocol is correct. The main point to improve is the definition of the planned slices. Relation type, entity degree, symmetry, inversion, and composition should be computed using only the training graph. Otherwise, the analysis may indirectly use validation/test information.

Also clarify whether model selection is done only on validation MRR and whether the test set is evaluated only once after final configuration. This is important if several sampling strategies and embedding dimensions will be tried.

4. Baseline and preliminary results

The baseline implementation is strong for Milestone 1. TransE and RotatE are appropriate references, and the preliminary results are plausible: RotatE substantially outperforms TransE across all reported metrics.

However, the baseline description should be more explicit. “Default per model in PyKEEN” is not enough for reproducibility. Specify the exact loss, margin if applicable, number of negatives per positive, negative sampler, embedding initialization, regularization, evaluator settings, and stopping criterion. If the comparison is described as “matched hyperparameters,” then the report should say exactly which parameters are matched and which are model-specific.

The 15-epoch results are useful, but they should be treated as preliminary. Since both models were still improving, Milestone 2 should report longer runs before drawing final conclusions.

5. Error analysis and next experiments

The planned error analysis is good, but it should be made concrete. For example, for each relation slice, report MRR/Hits@10 for TransE, RotatE, and RotatE plus hard negatives. Then show a few qualitative examples where RotatE fixes a TransE failure, and a few cases where both fail.

The relation-structural question is the most interesting part of the project. Define the categories explicitly: what counts as symmetric, inverse, compositional, one-to-many, many-to-one, and high-degree. Without these definitions, the analysis may become impressionistic.

6. Scope and feasibility

The project is well scoped. The minimum path is clear: full TransE/RotatE runs, hard-negative ablation, slice metrics, and error analysis. Avoid adding many more KGE models unless the main analysis is already complete. A clean comparison of two models plus one sampling intervention is stronger than a shallow benchmark of many models.

The custom hard-negative sampler is a reasonable contribution, but keep it modest. The goal should be to test whether harder negatives change aggregate performance and whether gains concentrate in particular relation or degree regimes.

Minimum viable final project

A successful minimum version of this project would be:

- Full reproducible TransE and RotatE runs on FB15k-237 with filtered MRR and Hits@k.
- A clear hard-negative sampling implementation and ablation against the default sampler.
- Relation-wise and degree-wise slice evaluation computed from the training graph.
- Training curves and validation curves for the final runs.
- Qualitative examples of successes and failures.

Optional extensions, only if the main experiments are stable:

- Embedding dimension ablation.
- More negative sampling variants.
- Comparison with one additional KGE model.
- Runtime and memory trade-off analysis.

Most important changes before Milestone 2

- Run longer training experiments and report final validation/test results.
- Specify all PyKEEN defaults that affect reproducibility.

- Define the structural relation categories precisely.
- Compute slice statistics using only the training graph.
- Add qualitative error analysis for representative relations and entities.

Grade

19

The priority now should be to make the interpretability analysis rigorous. The baseline is already working; the value of the final project will come from showing where and why RotatE improves over TransE, and whether hard-negative mining changes those gains in identifiable graph regimes.